ELECTRICAL ENGINEERING ASSET PORTFOLIO

Residential and Commercial Building Projects

(CAD Drawings, Schematics, Load Schedules and Technical Documentation)

Engineering Discipline: Electrical Engineering

Asset Categories:

- Electrical Power Systems
- Lighting Systems
- Extra Low Voltage (ELV) Systems
- Fire Alarm Systems
- CCTV Systems
- Load Schedules
- Consumer Unit (CU) Configurations

Software Used:

- AutoCAD
- Microsoft Excel

Document Version: Rev. 00

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Table of Contents

1. Portfolio Overview
2. Project 1 – Fire Alarm & CCTV System Design (5-Storey Hotel)
3. Project 2 – Lighting & Power Design (Residential Apartment)
4. Project 3 – Lighting, Power & ELV Design (2-Storey Residential Complex)
5. Project 4 – Power, Fire Alarm & CCTV Design (Commercial Complex)
6. Project 5 – TAN Residence
7. Project 6 – Terrace Building

8. Portfolio Summary
9. Professional Skills Demonstrated

Portfolio Overview

Introduction

This portfolio presents a collection of electrical engineering design assets developed for residential and commercial building projects. The assets include electrical power layouts, lighting layouts, fire alarm systems, CCTV systems, consumer unit arrangements, schematic diagrams, and electrical load schedules.

The drawings were developed using AutoCAD, while load calculations and schedules were prepared using Microsoft Excel where applicable.

The portfolio demonstrates practical experience in producing structured engineering documentation suitable for electrical installation, contractor reference, design review, and technical evaluation.

Purpose of this Portfolio

This portfolio has been compiled to:

- Demonstrate competency in electrical engineering design.
- Present professionally prepared CAD drawings and supporting documentation.
- Showcase electrical power and ELV system planning.
- Provide structured engineering assets for technical review.
- Illustrate project documentation and organization practices.

Asset Categories Included

- Electrical Power Layouts
- Lighting Layouts
- Fire Alarm Layouts
- Fire Alarm Schematics
- CCTV Layouts
- Consumer Unit (CU) Arrangements
- Circuit Distribution Drawings

- Electrical Load Schedules

PROJECT 1

Fire Alarm and CCTV System Design – 5-Storey Hotel

Asset Metadata

Item	Description
Project ID	P001
Building Type	Commercial Hotel
Engineering Discipline	Electrical Engineering
Asset Category	ELV Systems
Software	AutoCAD
File Format	DWG
Drawing Status	Completed Design

Project Overview

This project covers the design and drafting of the fire alarm and CCTV systems for a five-storey hotel. The objective was to provide effective fire detection, alarm notification, and surveillance coverage throughout the facility while maintaining an organized and maintainable system layout.

Scope of Work

- Fire alarm layout design for all floors
- CCTV camera layout design
- Fire alarm schematic development
- Consumer Unit (CU) distribution arrangement
- Fire detection zoning

- Device positioning based on architectural layouts

Engineering Assets Produced

- Fire Alarm Layout Plans
- CCTV Layout Plans
- Fire Alarm and CCTV System Schematics
- Consumer Unit Distribution Diagram

Deliverables

- AutoCAD Drawings (.dwg)

Design Considerations

The fire alarm system was developed with emphasis on detector spacing, manual call point accessibility, zoning strategy, and sounder coverage.

The CCTV system was arranged to provide surveillance of entrances, corridors, staircases, lobby areas, and other strategic locations while maintaining efficient cable routing.

Repository Structure

P001_5Storey_Hotel/

├── CAD_Drawing

└── README.md

Skills Demonstrated

- Commercial electrical drafting
- ELV system planning
- Fire alarm schematic preparation
- CCTV layout planning
- Multi-floor system coordination

PROJECT 2

Lighting and Power Design – Residential Apartment

Asset Metadata

Item	Description
Project ID	P002
Building Type	Residential Apartment
Engineering Discipline	Electrical Engineering
Asset Category	Lighting & Power
Software	AutoCAD, Microsoft Excel
File Formats	DWG, XLSX

Project Overview

This project presents the complete lighting and power distribution design for a residential apartment. The work includes electrical layouts, load calculations, circuit planning, and consumer unit configuration to ensure safe and balanced electrical distribution.

Scope of Work

- Lighting layout design
- Power layout design
- Load calculations
- Circuit distribution planning
- Load schedule preparation

Engineering Assets Produced

- Lighting Layout Plan
- Power Layout Plan

- Electrical Load Schedule

Drawing Inventory

Drawing No.	Description	File Name	Format
L-001	Lighting Layout	Lighting layout for residential apartment	DWG
P-001	Power Layout	Power layout for residential apartment	DWG
LS-001	Load Schedule	DB for residential apartment	XLSX

Deliverables

- AutoCAD Drawings (.dwg)
- Excel Load Schedule (.xlsx)

Design Considerations

The lighting system was arranged to achieve balanced illumination and logical circuit grouping. Power distribution was designed to maintain safe circuit loading, proper socket distribution, and effective consumer unit organization. Electrical calculations were used to determine connected load, demand load, protective device ratings, and cable sizing requirements.

Repository Structure

P002_Residential_Apartment/

├── Load_Schedules

├── CAD_Drawings

└── README.md

Skills Demonstrated

- Residential electrical design

- Lighting layout planning
- Load calculation
- Circuit planning
- Consumer unit configuration

PROJECT 3

Lighting, Power and ELV Design – 2-Storey Residential Complex

Asset Metadata

Item	Description
Project ID	P003
Building Type	2-Storey Residential Complex
Engineering Discipline	Electrical Engineering
Asset Category	Lighting, Power & ELV Systems
Software	AutoCAD
File Formats	DWG
Drawing Status	Completed Design

Project Overview

This project involved the complete electrical and Extra Low Voltage (ELV) system design for a two-storey residential complex. The design integrates lighting, power distribution, fire alarm, CCTV, and load scheduling to provide a coordinated electrical installation suitable for residential occupancy.

Scope of Work

- Lighting layout design for all floors
- Power distribution layout design

- Fire alarm layout design
- Fire alarm schematic preparation
- CCTV system layout design
- Load calculation and schedule preparation
- Circuit distribution planning
- Consumer Unit (CU) arrangement

Engineering Assets Produced

- Lighting Layout Plans
- Power Layout Plans
- Fire Alarm Layout Plans
- CCTV Layout Plans
- Electrical Load Schedule
- Circuit Distribution Schedule

Drawing Inventory

Drawing No.	Description	File Name	Format
L-001	Lighting Layout Plans	Lighting for 2-Storey Residential Complex	DWG
P-001	Power Layout Plans	Power for 2-Storey Residential Complex	DWG
ELV-001	Extra Low Voltage Plans	ELV for 2-Storey Residential Complex	DWG
LS-001	Load Schedule Plan	DB for 2-Storey Residential Complex	XLSX

Deliverables

- AutoCAD Drawings (.dwg)
- Microsoft Excel Load Schedule (.xlsx)

Design Considerations

The lighting system was designed to provide balanced illumination, efficient fixture placement, and logical circuit grouping throughout the building.

The power distribution system was planned to ensure safe circuit loading, balanced distribution, appropriate protective device selection, and efficient cable routing.

The ELV systems were coordinated to provide effective fire detection, alarm notification, and CCTV surveillance with organized cable pathways and maintainable system layouts.

Electrical load calculations were completed to establish connected load, estimated demand load, circuit breaker selection, cable sizing basis, and circuit load balancing.

Repository Structure

P003_Residential_Complex/

├── Load Schedules

----- CAD_Drawings

└── README.md

Skills Demonstrated

- Integrated electrical system design
- Residential power distribution planning
- ELV system coordination
- Fire alarm schematic preparation
- CCTV layout planning
- Load calculation and circuit balancing
- Professional engineering documentation

PROJECT 4

Lighting, Power, Fire Alarm and CCTV Design – Commercial Complex

Asset Metadata

Item	Description
Project ID	P004
Building Type	Commercial Complex
Engineering Discipline	Electrical Engineering
Asset Category	Power Distribution & ELV Systems
Software	AutoCAD, Microsoft Excel
File Formats	DWG, XLSX
Drawing Status	Completed Design

Project Overview

This project presents the electrical power distribution and Extra Low Voltage (ELV) system design for a commercial complex. The design includes power layouts, fire alarm systems, CCTV layouts, and electrical load documentation to support safe and reliable building operation.

Scope of Work

- Power layout design
- Lighting layout design
- Circuit distribution planning
- Fire alarm layout preparation
- Fire alarm schematic development
- CCTV layout design

- Load calculation
- Load schedule preparation
- Consumer Unit configuration

Engineering Assets Produced

- Power Layout Plans
- Lighting layout plans
- Circuit Distribution Drawings
- Fire Alarm Layout Plans
- Fire Alarm System Schematics
- CCTV Layout Plans
- Electrical Load Schedule

Drawing Inventory

Drawing No.	Description	File Name	Format
L-001	Lighting Layout Plans	Lighting for Commercial Complex	DWG
P-001	Power Layout Plans	Power for Commercial Complex	DWG
SCH-001	Fire Alarm And CCTV Schematic	ELV for Commercial Complex	DWG
LS-001	Load Schedule	DB for Commercial Complex	XLSX

Deliverables

- AutoCAD Drawings (.dwg)
- Microsoft Excel Load Schedule (.xlsx)

Design Considerations

The electrical power distribution was planned to provide balanced circuit loading, organised feeder routing, appropriate protective device selection, and efficient equipment supply arrangements.

The fire alarm system incorporates detector positioning, manual call point placement, system zoning, and notification device coverage to support reliable detection and maintenance.

The CCTV layout provides surveillance coverage of key circulation areas, entrances, and operational spaces with structured cable routing for installation and future servicing.

Electrical load calculations established connected load, demand load estimation, protective device sizing, and circuit distribution requirements.

Repository Structure

P004_Commercial_Complex/

├── Load_Schedules

├── CAD_Drawings

└── README.md

Skills Demonstrated

- Commercial electrical drafting
- Power distribution planning
- ELV system integration
- Fire alarm schematic preparation
- CCTV system planning
- Load analysis
- Engineering documentation

PROJECT 5

TAN Residence – Power, Lighting, Fire Alarm, CCTV and Consumer Unit Design

Asset Metadata

Item	Description
Project ID	P005
Building Type	Residential Building
Engineering Discipline	Electrical Engineering
Asset Category	Lighting, Power & ELV Systems
Software	AutoCAD
File Formats	DWG
Drawing Status	Completed Design

Project Overview

This project covers the electrical power, lighting, and Extra Low Voltage (ELV) system design for the TAN Residence. The design integrates lighting layouts, power distribution, fire alarm systems, CCTV layouts, and Consumer Unit arrangements into a coordinated residential electrical design package.

Scope of Work

- Lighting layout preparation
- Power distribution layout design
- Fire alarm layout preparation
- CCTV layout design
- Consumer Unit configuration

Engineering Assets Produced

- Lighting Layout Plans
- Power Layout Plans
- Fire Alarm Layout Plans

- CCTV Layout Plans
- Consumer Unit Arrangement

Drawing Inventory

Drawing No.	Description	File Name	Format
L-001	Lighting Layout	Lighting for TAN Residence	DWG
P-001	Power Layout	Power for TAN Residence	DWG
SCH-001	Fire Alarm And CCTV Shematics	ELV for TAN Residence	DWG

Deliverables

- **AutoCAD Drawings (.dwg)**

Design Considerations

The lighting layout was developed to provide balanced illumination and practical switching arrangements throughout the residence.

The electrical power system provides organised socket outlet distribution, dedicated equipment supply points, and a structured Consumer Unit arrangement for safe circuit management.

The fire alarm and CCTV layouts were coordinated to provide effective safety monitoring, surveillance coverage, and maintainable cable routing throughout the property.

Repository Structure

P005_TAN_Residence/

├── CAD_Drawings

└── README.md

Skills Demonstrated

- Residential electrical design
- Consumer Unit configuration
- Lighting layout planning
- ELV system coordination
- CCTV layout preparation
- Fire alarm system planning
- Professional drafting

PROJECT 6

Terrace Building – Power, Lighting, Fire Alarm and CCTV Design

Asset Metadata

Item	Description
Project ID	P006
Building Type	Residential Terrace Building
Engineering Discipline	Electrical Engineering
Asset Category	Lighting, Power & ELV Systems
Software	AutoCAD, Microsoft Excel
File Formats	DWG, XLSX
Drawing Status	Completed Design

Project Overview

This project presents the electrical design documentation for a residential terrace building, integrating lighting, power distribution, fire alarm systems, CCTV layouts, and electrical load schedules into a coordinated engineering package suitable for residential installation.

Scope of Work

- Lighting layout design
- Power layout design
- Fire alarm layout preparation
- CCTV layout design
- Load calculation
- Load schedule preparation
- Circuit distribution planning

Engineering Assets Produced

- Lighting Layout Plans
- Power Layout Plans
- Fire Alarm Layout Plans
- CCTV Layout Plans
- Electrical Load Schedule

Drawing Inventory

Drawing No.	Description	File Name	Format
L-001	Lighting Layout Plans	Lighting for TERRACE BUILDING	DWG
P-001	Power Layout Plans	Power for TERRACE BUILDING	DWG
SCH-001	Fire Alarm And CCTV Schematics Layout	ELV for TERRACE BUILDING	DWG
LS-001	Load Schedule	TERRACE BUILDING DB	XLSX

Deliverables

- AutoCAD Drawings (.dwg)
- Microsoft Excel Load Schedule (.xlsx)

Design Considerations

The lighting design was developed to provide appropriate illumination levels and balanced circuit grouping across the building.

The power distribution system was planned to achieve safe socket outlet distribution, balanced circuit loading, appropriate protective device selection, and organised cable routing.

The fire alarm and CCTV systems were coordinated to ensure effective detector placement, surveillance coverage, maintainable cable routing, and seamless integration with the overall electrical installation.

Electrical load calculations established connected load, estimated demand, protective device selection criteria, cable sizing basis, and circuit balancing requirements.

Repository Structure

P006_Terrace_Building/

├── CAD_Drawings

├── README.md

└── Load_Schedules

Skills Demonstrated

- Residential electrical design
- Lighting and power distribution planning
- ELV system coordination
- Load calculation and schedule preparation
- Circuit distribution planning
- Professional engineering documentation

PROJECT SUMMARY

Above are selected engineering projects carefully done in accordance with engineering standards. I believe they would be of great help in improving AI systems.